

Inaccessibility to 911 and 311 Call Log Data and the Limitations it Creates for AI-Driven Hazard Detection: Case Studies of NYC Floods (2025)

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Urban flood hazards have been increasingly analyzed by AI softwares trained to pull information from a variety of sources, most of which are news articles, NOAA forecasts, and other external sources. However, despite the good documentation of floodings and the widespread impacts that these sources cover, there is little to no incorporation of individual flood related calls that are logged in 311 and 911 datasets in AI outputs. Unfortunately, most AI-softwares do not have access to these databases in the first place. Oftentimes, this data is incredibly inaccessible without human aid in decoding, filtering, and feeding the relevant information to them. As a result, when asked to comment on trends in NYC flooding for a specific year, their responses are almost entirely based on major news articles. This creates limitations in AI-software's ability to accurately determine instances of flooding in NYC for a specific timeframe, as it will only be able to confidently speak on major floods while ignoring instances that did not make headlines. This leads to inaccuracies in analyzing flood trends across NYC, stating them to occur much less frequently than they actually do. It is especially important for these trends to be adequately recorded as climate change and global warming continue to impact current flood rates. We aimed to explore these limitations by asking three common AI-softwares (ChatGPT, Perplexity, and Microsoft CoPilot) of their existing knowledge of NYC floods and their patterns for the year of 2025, and reviewed where they pulled their information. We also explored where they pulled their information on two recent major flood events, October 30th and July 14th 2025. Then, we exported recent datasets from NYC OpenData, and analyzed ~7 million 911 incident records, along with ~6 million 311 incident records all from 2025. We used RStudio and ChatGPT to filter the datasets and make them readable. We narrowed the datasets to show us the calls made on July 14th and October 30th, in addition to any other calls marked as flooding. Data was filtered by date, borough, and keywords (e.g., "Water," "Sewer," "Rescue," "Flood"), to locate exactly how many calls were made in relation to these floods. We found that these calls are not present within 911 databases, and most of them are logged under 311 databases. After feeding the filtered data to AI-Softwares, they were more confident in noting trends, patterns, and instances of flooding in 2025 when they had real-time data to cross reference with their original sources. This emphasizes the importance of integrating multi-source data and improving the accessibility of these datasets to AI-Software responsible for making decisions about flooding in NYC. It would prove useful to avoid inaccurate AI-driven decisions and ensure the most accurate information is being used in pattern detection.

LIMITATIONS

- Only worked with 3 of the more popular AI companions, ChaptGPT, Microsoft Copilot, and and there is a tendency for AI-bias that causes them to agree with what you are telling them
- It was incredibly time consuming to filter data and took extensive coding to make readable and feed it to AI
- There are little to no flooding calls logged under NYC's 911 call logs.
- NYC OpenData Datasets are not published immediately, often takes months to a year for it to be exportable

Chat GPT

- Most recent knowledge of NYC flooding was Oct 30 and July 14
- Says flooding is caused by sealevel rise, intense rainfall, and sewer backup, flashfloods are the biggest threat
- Information is pulled from news sources such as NYT, ABC, CBS, NOAA forecasts, youtube videos, and other articles
- Does not use 311/911 logs due to inaccessibility, messiness, and delays in availability to public
- Expresses that these logs would be useful in combination with other sources for pattern detection
- Worked with the dataset well, was able to analyze data when it was filtered and fed to them manually

Perplexity

- Most recent knowledge of NYC flooding was Oct 30 and July 31
- Flooding is caused by intense rainfall, flashflooding, drainage incapacity, rainfall rates are high for 2025
- Information pulled from news sources such as NYtimes, NOAA reports, FOX weather, or blogs
- Does not use 311/911 logs due to restriction issues to raw data, only has access to public summaries of them
- Expresses that these logs would make them feel more confident in pattern and trend detection
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Microsoft CoPilot

- Most recent knowledge of NYC flooding was July 14 and Oct 30
- Flooding is caused by localized rainfall intensity, lack of drainage capacity, and high tide floodings,
- Information is pulled from Previsco, FOX weather, NY Sea Grant, and other post flooding articles
- Does not use 311/911 logs due to public restrictions, cannot see the data unless it is cited in news sources, and the size of unfiltered datasets
- Expresses that it can read the data if it had access to it, and that it is useful in its ability to determine patterns and frequencies of calls that are not seen in articles
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THINGS TO DO

- Filter all flood data for 2025 in R, feed to them as a CSV
- Examine any new information they pull from it

However, these datasets often focus on encoding the details of flood responses among multiple broader categories, rather than its own category. This causes flood related calls and their emergency responses to be unspecified and hard to locate within the dataset without the use of external coding. This raises concerns for both scientific interpretation and emerging AI systems that rely on such data for decision-making and synthesizing information. Despite good documentation of the floodings' fatalities and widespread impacts in news articles, very few incidents in the police dataset were explicitly labeled as water-related across both events. Few incidents where water-rescue responses were reported, were geolocated outside the affected perimeter and were recorded at times where flooding was not present. However, temporal analysis on the 911 calls of July 14th and October 30th revealed consistent surges in calls beginning ~3:30–4:00 PM, aligning with independently reported flood onset. This indicates that flood related responses were not formally recorded in the dataset. These results demonstrate a systematic underrepresentation of flood hazards in police dispatch data, where indications of the floods taking place are fragmented across unrelated categories such as "time", "response duration", "outside". Flood events are not formally classified in an organized category that reflects the information that would typically be grouped together in any other kind of emergency.

This poses a critical risk for AI and big-data systems trained on such data, to misclassify hazard severity and underestimate emergency demand. Police dispatch data alone are insufficient for direct hazard detection but can reveal event dynamics through indirect indications. Integrating multi-source data and improving classification frameworks among dispatch datasets are essential to avoid inaccurate AI-driven decisions and ensure adequate emergency response funding.

[311 Service Requests from 2020 to Present | NYC Open Data](#)